North American Bat Monitoring Program in Mount Revelstoke and Glacier National Parks

Manual Verification Data Summary 2025

Camila Hurtado Brooke Daly

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1 Executive Summary

2 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on Treaty 8 and Douglas Treaties Territories as well as the traditional and unceded territories of First Nations and Métis Peoples across all regions of British Columbia, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

3 Background

3.1 Overview of NABat and the NNW Bat Hub

The North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species across North America using standardized protocols and a unified sample design (Loeb et al. 2015). NABat was established to address the gaps in knowledge and lack of long-term studies of bat species across Mexico, USA, and Canada. The program is administered by the US Geological Survey (USGS), coordinated by the Canadian Wildlife Health Cooperative (CWHC) in Canada, and implemented by the North by Northwest Bat (NNW) Hub in British Columbia, Alberta, and S.E. Alaska.

3.2 NABat Motioning at Mount Revelstoke and Glacier National Parks

Mount Revelstoke and Glacier National Parks have monitored two NABat grid cells since 2017 and have collected 9 years of data. Both grid cells are in southeastern BC. Mount Revelstoke contains

2 stationary bat detectors and 1 transect route while Glacier contains 3 stationary detectors, all detectors were deployed in primarily forest habitat.

4 Methodology

Full-spectrum recordings from the sampling periods were processed using two automatic classifiers: Kaleidoscope's Bats of North America 5.4.0 classifier and Sonobat 3.0's Southeastern British Columbia classifier. Based on documented species ranges, manual identification efforts focused on 10 species: Townsend's big-eared bat (*Corynorhinus townsendii*), Big Brown Bats (*Eptesicus fuscus*), Silver-haired Bats (*Lasionycteris noctivagans*), Hoary Bat (*Lasiurus cinereus*), California Myotis (*Myotis californicus*), Long-eared myotis (*Myotis evotis*), Little Brown Myotis (*Myotis lucifugus*), Northen Myotis (*Myotis spententrionalis*), Long-legged Myotis (*Myotis volans*), and Yuma Myotis (*Myotis yumanensis*).

The analysis workflow followed processing standards established by the North American Bat Monitoring Program (NABat) (Reichert et al. 2018). Only recordings that were identified as a bat by Kaleidoscope were manually verified. Species identifications were validated using reference call parameters described by Szewczak (2018), Slough et al. (2022), and Solick (2022), in accordance with NABat manual vetting protocols (Reichert et al. 2018).

5 Results and Discussion

All previously detected species were identified again across sites.

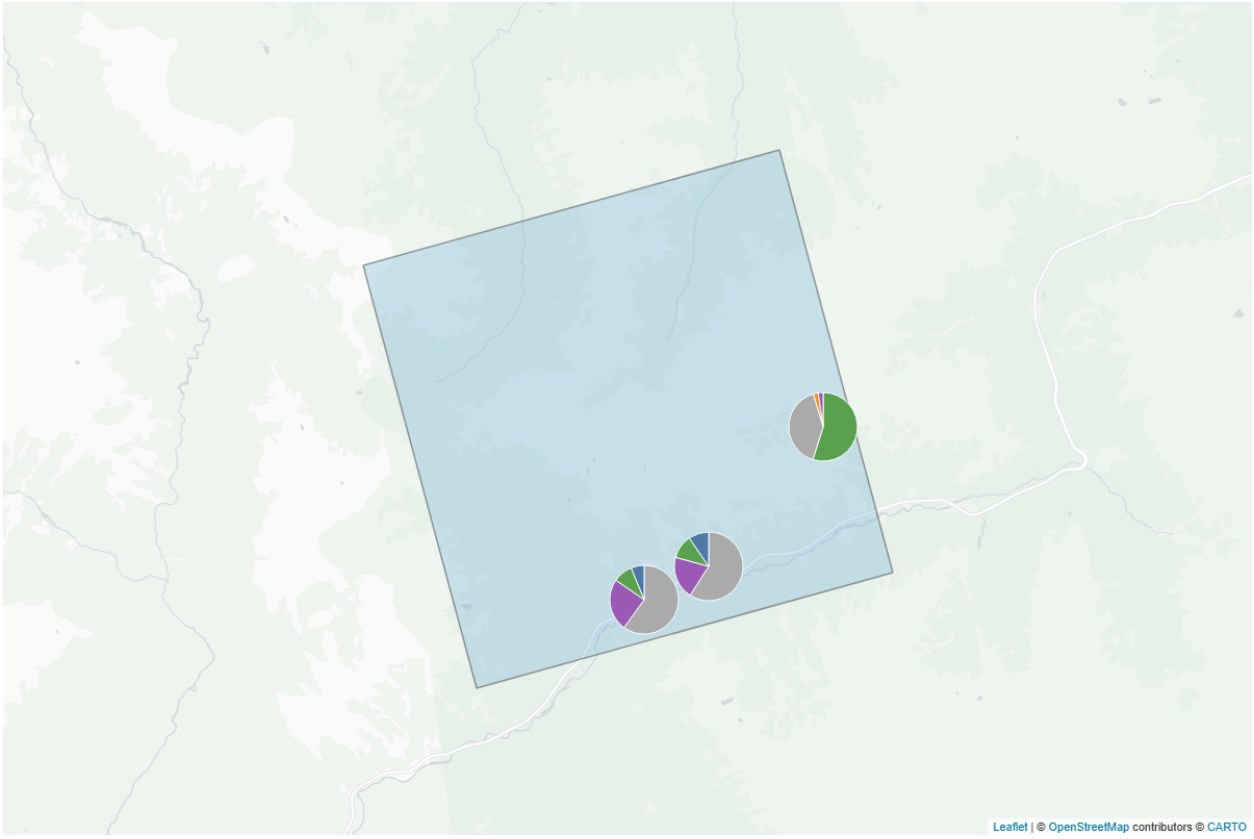

Figure 1: Sampled sites in Glacier National Park (GRTS 271731). Proportion of species bat passes at each site. Other represents tags that were identified as species couplets or species groups

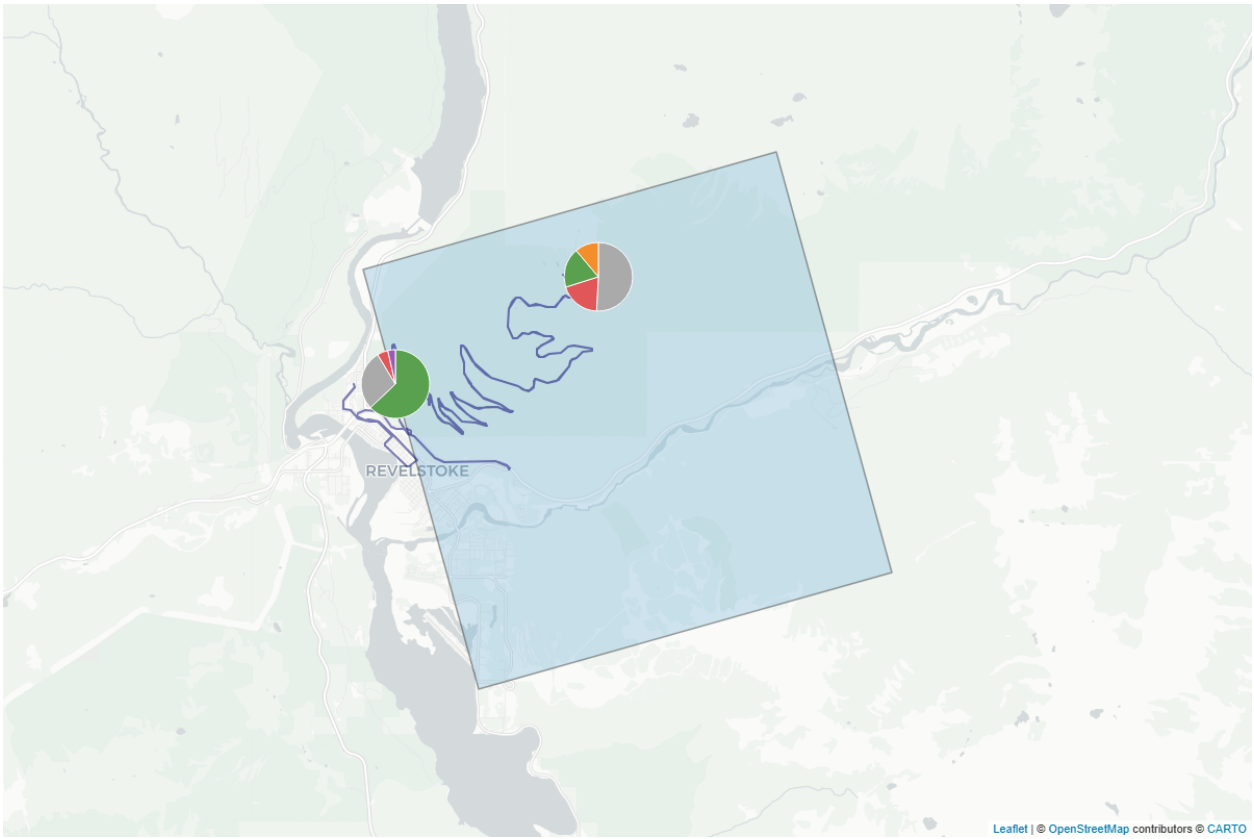

Figure 2: Sampled sites and driving transect route in Mount Revelstoke National Park (GRTS 241002). Proportion of species bat passes at each site. Other represents tags that were identified as species couplets or species groups

All tag data has been uploaded to NABat under project ID: . Recordings and tag data can also be found in Wildtrax

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